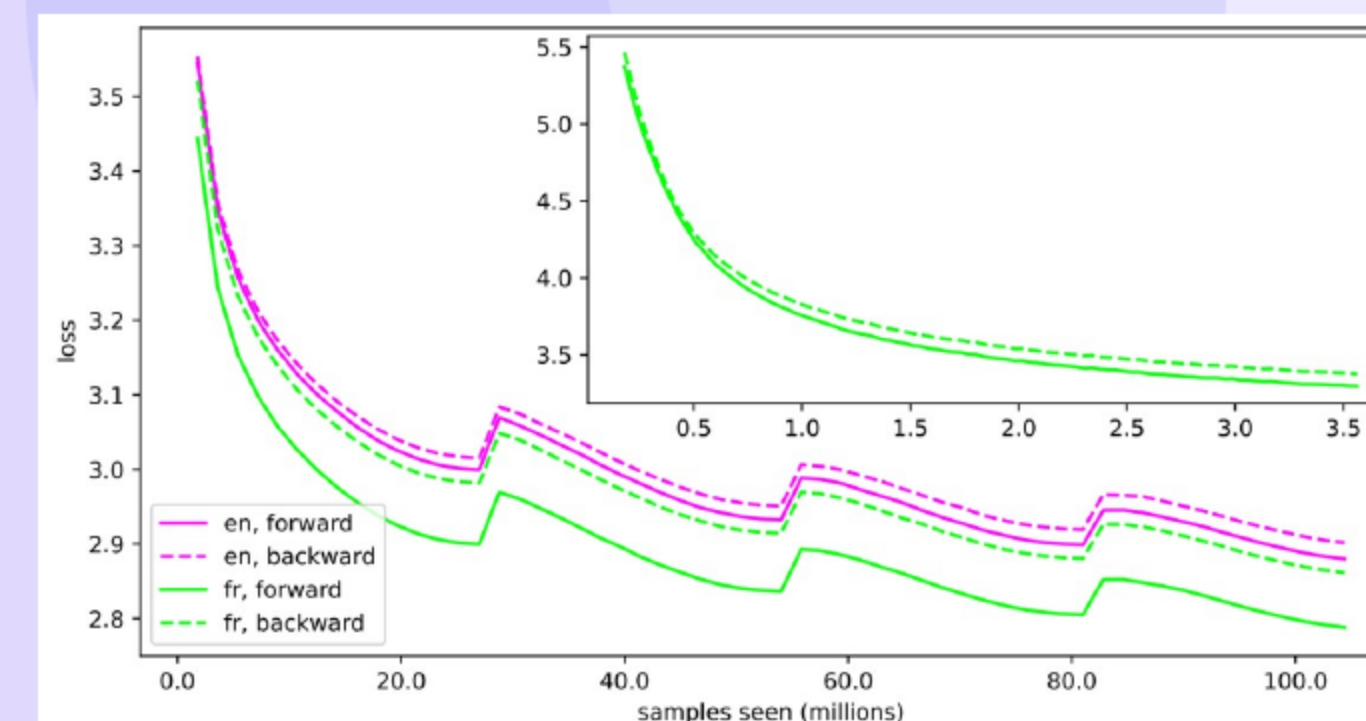


Arrows of Time for Large Language Models

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01 Visual Intuition: Forward vs Backward Prediction in LLMs



- We illustrate forward ($\rightarrow$) vs backward ($\leftarrow$) prediction as context grows.
- We find backward loss $>$ forward loss for large models and long contexts.

02 Empirical Findings and Mechanisms

- We show a robust forward advantage across languages, sizes, contexts, and RNNs; the gap grows with context and scale.
- We rule out tokenizer and schedule artifacts.
- We explain AoT via asymmetric computability/sparsity: multiplication vs factoring and sparse maps with dense inverses.

03 Conclusion and Takeaway

- We conclude forward prediction is systematically easier, and the gap widens with model capacity and context.
- We attribute this to computational hardness and sparsity asymmetries.
- We outline open directions in other modalities and scaling.

Thank You

Questions & Discussion